Table Of Contents

Encrypt Data fitting 2
Encrypt Data fitting example 4
Example: Extract a list and encrypt and upload to SFTP (mock test) 9
Example: Extract a list and encrypt and upload to SFTP 19

Encrypt Data fitting

The Encrypt Data fitting supports PGP encryption with digital signing for data integrity and authenticity. It encrypts incoming data and returns encrypted data as binary (or ASCII armored, if specified).

Expected input

Input to the fitting must be in a single payload.

The Encrypt Data fitting supports multiple input types.

Type	Description
String	Text data to be encrypted.
Buffer/Uint8Array	Binary data to be encrypted.
Array of strings	Reducer fitting output of multiple text payloads.
Array of Buffers/Uint8Arrays	Reducer fitting output of multiple binary payloads.

Reference configuration

Configuration	Description
Source	Optional. String. JSONPath expression to extract data from the message payload for encryption. Defaults to entire payload.
Public Key	Required. String. Pipeline parameter name containing the PGP public key to use for encryption.
Signature	Optional. Boolean. If true, the data will be digitally signed using the sender's provided private key.

Configuration	Description
	Private Key. Required (String) if Signature is true. Pipeline parameter name containing the PGP private key to use for signing. **Passphrase**. Optional. String. Pipeline parameter name containing the passphrase for the Private Key (if password-protected).
Target	Optional. String. Dot-notation path to assign the encrypted output in the message payload. Defaults to replacing the entire payload.
Binary	Optional. Boolean. If true, returns binary encrypted data. If false, returns ASCII-armored format.
Ignore Errors	Optional. Boolean. If true, continues processing on error and routes to errorPath.
Error Path	Required if Ignore Errors is true. String. Dot-notation path for placing error info in the message.

Size limits

Maximum data size for fitting input is 10MB.

Output

Binary format (default): Encrypted data returned as Buffer when `binary: true`.

ASCII armored: Encrypted data returned as ASCII armored when `binary: false`.

```
-----BEGIN PGP MESSAGE-----
[encrypted content]
-----END PGP MESSAGE-----
```

Usage examples

Example fitting configuration for basic encryption:

```
{
  "class": "encryptData",
  "config": {
    "publicKey": "recipientPublicKey"
  }
}
```

Example of extracting data from a specific path and outputting as binary to pass down to fittings such as SFTP Put, File Writer:

```
{
  "class": "encryptData",
  "config": {
    "publicKey": "recipientPublicKey",
    "source": "$.sensitiveData",
"target": "payload.encryptedData",
    "binary": true
  }
}
```

Example of how to sign a file using a passphrase and private key, and continue pipeline processing even if an error occurs:

```
{
  "class": "encryptData",
  "config": {
    "publicKey": "recipientPublicKey",
    "signature": true,
    "signingKey": {
      "privateKey": "signerPrivateKey",
"passphrase": "keyPassphrase"
    },
    "ignoreErrors": true,
    "errorPath": "payload.encryptionErrors"
  }
}
```

Encrypt Data fitting example

This example provides sample pipelines demonstrating how to use the Encrypt Data fitting in Data Connect to apply Pretty Good Privacy (PGP) encryption to sensitive data before transferring it to external systems.

The Encrypt Data fitting enables secure, standards-compliant encryption of data within your pipelines. It supports (PGP) encryption with optional digital signatures, making it ideal for protecting sensitive information during data transfers to external partners, government agencies, or third-party systems.

In this example, the pipeline extracts student worker records from Banner, formats the data as CSV, encrypts it using the Encrypt Data fitting with PGP, and uploads the encrypted file to an SFTP server.

For example, a Human Resources manager that has student worker data that is stored in Banner. As an HR admin, you want to make sure that the data is encrypted every time you extract it from Banner, so that it can be safely transmitted for various purposes. Then a pipeline developer who is part of the security team wants to make sure that they are doing that encryption safely before transmitting that information.

The person who receives that encrypted student worker data would use a different pipeline with the Decrypt Data fitting.

Prerequisites

You will need the following before you begin:

- Experience Premium with Integration Designer
- Insights with the Extensibility license (for the Extract Data fitting)
- PGP key pair for encryption and signing (public key, private key, and passphrase)
- SFTP server credentials (private key, username, host, and destination path)

PGP encryption

The Encrypt Data fitting provides PGP encryption capabilities with the following features:

Type	Description
PGP Encryption	Encrypts data using a PGP public key, ensuring only the intended recipient can decrypt it.
Digital Signatures	Optionally signs the encrypted data with a private key to verify authenticity and integrity.
Binary Output	Produces binary encrypted output suitable for file transfer.
Error Handling	Configurable error handling with the Ignore Errors option.

Included pipelines

encrypt-student-worker-data-mock

This pipeline uses mock data to demonstrate the Encrypt Data fitting without requiring a real connection.

For detailed steps, see Example: Extract a list and encrypt and upload to SFTP (mock test).

encrypt-student-worker-data

This pipeline extracts real student worker data from Banner, encrypts it using the Encrypt Data fitting, and uploads it to SFTP.

For detailed steps, see Example: Extract a list and encrypt and upload to SFTP.

Parameters

publicKeyEncrypt	The PGP public key used to encrypt the data file.
privateKeyEncrypt	The PGP private key used to sign the encrypted file for authenticity verification.

passphraseEncrypt	The passphrase associated with the PGP

Parameter	Description
	private key used for signing.
privateKeySFTP	The private key used for authenticating with the SFTP server.
sftpUsername	The username for authenticating with the SFTP server.
sftpHost	The hostname or IP address of the SFTP server.
sftpFilePath	The full file path on the SFTP server where the encrypted file will be uploaded.

Execution instructions

Security considerations

The Encrypt Data fitting provides multiple layers of security:

- PGP Encryption: Uses asymmetric encryption where data is encrypted with a public key and can only be decrypted with the corresponding private key held by the recipient.
- Digital Signatures: When signature: true, the fitting signs the data with your private key, allowing recipients to verify the data's authenticity and that it hasn't been tampered with.
- Sensitive Parameters: All encryption keys and credentials should be marked as sensitive: true in parameters to ensure they are stored securely and not logged.
- Secure Transfer: Combined with SFTP, provides end-to-end security for data in transit.

Other Use Cases for Encrypt Data Fitting

The Encrypt Data fitting is essential for scenarios requiring secure data transfer:

1. Payroll exports to external payroll providers

2. Financial data transfers to auditing firms
3. HR data synchronization with third-party HR systems

4. Student records exports for verification services
5. Vendor data exchanges requiring encryption compliance

File Output

The pipeline generates a PGP-encrypted .gpg file containing the student worker data in CSV format. The file is uploaded to the specified SFTP location with the configured filename.

Example: Extract a list and encrypt and upload to SFTP (mock test)

Create a pipeline that demonstrates the encryption and SFTP upload workflow using mock student worker data.

About this task

This pipeline is designed for testing and development purposes without requiring a database connection.
This test version of the pipeline:

- Generates synthetic student worker data using a JavaScript Transform fitting
- Demonstrates the full Encrypt Data fitting workflow with mock data
- Is useful for testing encryption configuration without affecting real data

Procedure

1. In Ellucian Experience, click the **Integration Designer** card.
2. Select or create a package:
    a. Select the Integration package where this new pipeline will be created.
    b. If needed, create a new package.
    c. In the **Packages** area, click **+**.
    d. Enter a unique name for the package.
    e. **Optional:** Add a description.

f. Click **Save**.

g. Select the new package from the list to continue.

3. Create a new pipeline:

   a. In the **Pipelines** area, click **+ PIPELINE**.

   b. Enter a unique name for the pipeline (for example, `Encrypt Student Worker Data Pipeline Mock`).

   c. **Optional:** Add a description. For example, `This pipeline demonstrates the encryption and SFTP upload workflow using mock student worker data. It is designed for testing and development purposes without requiring a database connection.`

   d. Click **Save**.

   e. Click the pipeline name to open it.

   f. To set up parameters, near the pipeline name, click the drop-down menu and select **Edit Pipeline Settings**.

   g. Add the following new pipeline parameters:

      a. publicKeyEncrypt

         - Label: `PGP Public Key`
         - Description: `The PGP public key used to encrypt the student worker data file.`
         - Type: **String**
         - Required: **True**
         - Sensitive: **True**
         - Multiline: **True**
         - Persistent: **True**

      b. privateKeyEncrypt

         - Label: `PGP Private Key for Signing`

- Description: `The PGP private key used to sign the encrypted file for authenticity verification.`
- Type: **String**
- Required: **True**
- Sensitive: **True**
- Multiline: **True**
- Persistent: **True**

c. passphraseEncrypt

- Label: `PGP Passphrase`
- Description: `The passphrase associated with the PGP private key used for signing.`
- Type: **String**
- Required: **True**
- Sensitive: **True**
- Persistent: **True**

d. privateKeySFTP

- Label: `SFTP Private Key`
- Description: `The private key used for authenticating with the SFTP server.`
- Type: **String**
- Required: **True**
- Sensitive: **True**
- Multiline: **True**
- Persistent: **True**

e. sftpUsername

- Label: `SFTP Username`
- Description: `The username for authenticating with the SFTP server.`
- Type: **String**
- Required: **True**

f. sftpHost

- Label: `SFTP Host`
- Description: `The hostname or IP address of the SFTP server.`
- Type: **String**
- Required: **True**

g. sftpFilePath

- Label: `SFTP Destination File Path`
- Description: `The full file path on the SFTP server where the encrypted file will be uploaded.`
- Type: **String**
- Required: **True**

h. Click **Save**.

4. To build the first segment for this example, configure the JavaScript Transform fitting:

   a. from the **Transformations** menu, click **JavaScript Transform**.

   b. Enter `Generate Mock Student Worker Data` as the fitting name.

   c. **Optional:** Enter a description. For example, `Generates mock student worker data for testing the encryption and SFTP upload workflow without requiring a database connection.`

   d. Set **Push Undefined** to **True**.

e. Set **Stop On Error** to **False**.

f. Enter the following:

```
"function transform(message, context) {
message.payload.studentWorkers = [
    {
      "spriden_id": "STU001",
      "spriden_first_name": "John",
      "spriden_last_name": "Smith",
      "nbrjobs_sal_grade": "A1",
      "nbrjobs_ann_salary": 15000,
      "nbrjobs_posn": "SW0001",
      "nbbposn_title": "Library Assistant",
      "nbrjobs_orgn_code_ts": "LIB01",
      "pebempl_orgn_code_home": "ACAD",
      "pebempl_orgn_code_dist": "MAIN",
      "nbbposn_grade": "01",
      "nbbposn_exempt_ind": "N",
      "nbbposn_posn_reports": "MGR001",
      "nbrjobs_suff": "00",
      "nbrjobs_ecls_code": "ST"
    },
    {
      "spriden_id": "STU002",
      "spriden_first_name": "Emily",
      "spriden_last_name": "Johnson",
      "nbrjobs_sal_grade": "A2",
      "nbrjobs_ann_salary": 16500,
      "nbrjobs_posn": "SW0002",
      "nbbposn_title": "Research Assistant",
      "nbrjobs_orgn_code_ts": "RES01",
      "pebempl_orgn_code_home": "GRAD",
      "pebempl_orgn_code_dist": "MAIN",
      "nbbposn_grade": "02",
      "nbbposn_exempt_ind": "N",
      "nbbposn_posn_reports": "MGR002",
      "nbrjobs_suff": "00",
      "nbrjobs_ecls_code": "US"
    },
    {
      "spriden_id": "STU003",
      "spriden_first_name": "Michael",
      "spriden_last_name": "Williams",
      "nbrjobs_sal_grade": "A1",
      "nbrjobs_ann_salary": 14500,
      "nbrjobs_posn": "SW0003",
      "nbbposn_title": "IT Help Desk",
      "nbrjobs_orgn_code_ts": "ITS01",
      "pebempl_orgn_code_home": "TECH",
```

```
  "pebempl_orgn_code_dist": "MAIN",
  "nbbposn_grade": "01",
  "nbbposn_exempt_ind": "N",
  "nbbposn_posn_reports": "MGR003",
  "nbrjobs_suff": "00",
"nbrjobs_ecls_code": "ST"
},
{
  "spriden_id": "STU004",
  "spriden_first_name": "Sarah",
  "spriden_last_name": "Davis",
  "nbrjobs_sal_grade": "B1",
  "nbrjobs_ann_salary": 18000,
  "nbrjobs_posn": "SW0004",
  "nbbposn_title": "Lab Technician",
  "nbrjobs_orgn_code_ts": "SCI01",
  "pebempl_orgn_code_home": "STEM",
  "pebempl_orgn_code_dist": "NORTH",
  "nbbposn_grade": "03",
  "nbbposn_exempt_ind": "N",
  "nbbposn_posn_reports": "MGR004",
  "nbrjobs_suff": "00",
  "nbrjobs_ecls_code": "GB"
},
{
  "spriden_id": "STU005",
  "spriden_first_name": "David",
  "spriden_last_name": "Brown",
  "nbrjobs_sal_grade": "A2",
  "nbrjobs_ann_salary": 15500,
  "nbrjobs_posn": "SW0005",
  "nbbposn_title": "Administrative Assistant",
  "nbrjobs_orgn_code_ts": "ADM01",
  "pebempl_orgn_code_home": "ADMIN",
  "pebempl_orgn_code_dist": "MAIN",
  "nbbposn_grade": "02",
  "nbbposn_exempt_ind": "N",
  "nbbposn_posn_reports": "MGR005",
  "nbrjobs_suff": "00",
  "nbrjobs_ecls_code": "NR"
},
{
  "spriden_id": "STU006",
  "spriden_first_name": "Jennifer",
  "spriden_last_name": "Martinez",
  "nbrjobs_sal_grade": "A1",
  "nbrjobs_ann_salary": 14000,
  "nbrjobs_posn": "SW0006",
  "nbbposn_title": "Tutoring Center Assistant",
  "nbrjobs_orgn_code_ts": "TUT01",
  "pebempl_orgn_code_home": "ACAD",
```

"pebempl_orgn_code_dist": "MAIN",
"nbbposn_grade": "01",
"nbbposn_exempt_ind": "N", "nbbposn_posn_reports": "MGR006",
"nbrjobs_suff": "00",
"nbrjobs_ecls_code": "ST"
},
{
"spriden_id": "STU007",
"spriden_first_name": "Christopher",
"spriden_last_name": "Garcia",
"nbrjobs_sal_grade": "B2",
"nbrjobs_ann_salary": 19000,
"nbrjobs_posn": "SW0007",
"nbbposn_title": "Graduate Teaching Assistant",
"nbrjobs_orgn_code_ts": "ENG01",
"pebempl_orgn_code_home": "GRAD",
"pebempl_orgn_code_dist": "SOUTH",
"nbbposn_grade": "04",
"nbbposn_exempt_ind": "Y",
"nbbposn_posn_reports": "MGR007",
"nbrjobs_suff": "00",
"nbrjobs_ecls_code": "GB"
},
{
"spriden_id": "STU008",
"spriden_first_name": "Amanda",
"spriden_last_name": "Wilson",
"nbrjobs_sal_grade": "A1",
"nbrjobs_ann_salary": 13500,
"nbrjobs_posn": "SW0008",
"nbbposn_title": "Recreation Center Staff",
"nbrjobs_orgn_code_ts": "REC01",
"pebempl_orgn_code_home": "ATHL",
"pebempl_orgn_code_dist": "MAIN",
"nbbposn_grade": "01",
"nbbposn_exempt_ind": "N",
"nbbposn_posn_reports": "MGR008",
"nbrjobs_suff": "00",
"nbrjobs_ecls_code": "ST"
},
{
"spriden_id": "STU009",
"spriden_first_name": "Daniel",
"spriden_last_name": "Anderson",
"nbrjobs_sal_grade": "A2",
"nbrjobs_ann_salary": 17000,
"nbrjobs_posn": "SW0009",
"nbbposn_title": "Media Services Assistant",
"nbrjobs_orgn_code_ts": "MED01",
"pebempl_orgn_code_home": "COMM",
"pebempl_orgn_code_dist": "MAIN",

```
            "nbbposn_grade": "02",
            "nbbposn_exempt_ind": "N",
            "nbbposn_posn_reports": "MGR009", "nbrjobs_suff": "00",
            "nbrjobs_ecls_code": "US"
          },
          {
            "spriden_id": "STU010",
            "spriden_first_name": "Jessica",
            "spriden_last_name": "Taylor",
            "nbrjobs_sal_grade": "A1",
            "nbrjobs_ann_salary": 14500,
            "nbrjobs_posn": "SW0010",
            "nbbposn_title": "Student Success Center Peer Mentor",
            "nbrjobs_orgn_code_ts": "SSC01",
            "pebempl_orgn_code_home": "STAF",
            "pebempl_orgn_code_dist": "MAIN",
            "nbbposn_grade": "01",
            "nbbposn_exempt_ind": "N",
            "nbbposn_posn_reports": "MGR010",
            "nbrjobs_suff": "00",
            "nbrjobs_ecls_code": "ST"
          }
        ];

          return message; }
```


g. Click **Test**.

5. To iterate student workers, configure the For Each fitting.

   a. From the **Transformations** menu, click **For Each**.

   b. Enter `Iterate Student Workers` as the fitting name.

   c. **Optional:** Enter a description. For example, `Iterates over each student worker record in the mock data collection.`

   d. Enter `payload.studentWorkers` in the **Source** field.

   e. Click **Test**.

6. To specify the path to where to extract the data from before you convert the data into a CSV file, configure the Delimited Formatter fitting:

   a. From the **Transformations** menu, select **Delimited Formatter**.

   b. Enter `Format Data as CSV` as the fitting name.

c. **Optional:** Enter a description. For example, `Formats the mock student worker data into a CSV structure with selected columns including name, salary grade, annual salary, position code, and title.`

d. Set **Normalize Columns** to **False**.

e. Turn **Column Headers True**.

Column paths from the Extract Data transform display in the **Columns** field.

f. Click **Test**.

Result: The data is now formatted to write to a CSV file.

7. To write that formatted data to the CSV file, configure the File Writer fitting:

   a. From the **Destinations** menu, select **File Writer**.

   b. Enter `Write CSV to Temporary Storage` as the fitting name.

   c. **Optional:** Enter a description. For example, `Writes the formatted CSV data to temporary S3 storage for subsequent encryption processing.`

   d. Enter a **File Name**. For example, `studentWorkers.csv`.

   e. Set **Add New Line After Each Message** to **True**.

   f. Set **Include Total Count** to **False**.

   g. Set **Generate Presigned Url** to **True**.

   h. Set **Write Immediately** to **False**.

   i. Set **UTF Encoding** to **UTF-8**.

   j. Click **Test**.

8. Configure the File Reader fitting:

   a. From the **Sources** menu, select **File Reader**.

   b. Enter `Read CSV File` as the fitting name.

   c. **Optional:** Enter a description. For example, `Reads the CSV file from S3 temporary storage to prepare it for encryption.` d. Set **Key** to

`{{context.__pipelineName}}/{{context.__runId}}/studentWorkers.csv`

. e. Set **Binary** to **False**.

f. Set **Allow Concurrent** to **False**.

g. Click **Test**.

9. Configure the Encrypt Data fitting:

   a. From the **Transformations** menu, select **Encrypt Data**.

   b. Enter `Encrypt Data with PGP` as the fitting name.

   c. **Optional:** Enter a description. For example, `Encrypts the CSV file content using PGP encryption with the provided public key and signs it with the private key for authenticity.`

   d. Select the **Public Key** parameter `publicKeyEncrypt` that you set up in the pipeline parameters earlier from the drop-down list.

   e. Set **Signature** to **True**.

      The **Private Key** becomes required.

   f. Enter `privateKeyEncrypt` for the **Private Key** and `passphraseEncrypt` for the **Passphrase**.

   g. Leave the **Binary** setting as the default setting **True** to return encrypted data in binary format.

   h. Set **Ignore Errors** to **False**.

   i. Click **Test**.

10. Configure the SFTP Put fitting:

    a. From the **Destinations** menu, select SFTP Put.

    b. Enter `Upload Encrypted File to SFTP` as the fitting name.

    c. **Optional:** Enter a description. For example, `Securely uploads the encrypted GPG file to the configured SFTP server using private key authentication.`

    d. Enter `privateKeySFTP` for the **Private Key**.

e. Enter `{{context.sftpUsername}}` for the **User Name**.

f. Enter `{{context.sftpHost}}` for the **Host**.

g. Enter `{{context.sftpFilePath}}` for the **File Path**.

h. Set **Push** to **True**.

i. Click **Test**.

Results

The pipeline generates a PGP-encrypted .gpg file containing the student worker data in CSV format. The file is uploaded to the specified SFTP location with the configured filename.

Example: Extract a list and encrypt and upload to SFTP

Create a pipeline that extracts student worker records from Banner, formats the data as CSV, encrypts it using the Encrypt Data fitting with PGP, and uploads the encrypted file to an SFTP server.

About this task

This is the production pipeline that performs the following steps:

- Extract Data: Queries Banner for active student worker records including employee details, position information, and salary data.
- Delimited Formatter: Formats the extracted data into a CSV structure with selected columns.
- File Writer: Stores the formatted CSV in S3 temporary storage.
- File Reader: Reads the CSV file from temporary storage for encryption.
- Encrypt Data: Encrypts the file using PGP encryption and signs it with a digital signature.
- SFTP Put: Securely uploads the encrypted GPG file to the configured SFTP server.

Procedure

1. In Ellucian Experience, click the **Integration Designer** card.

2. Select or create a package:
    a. Select the Integration package where this new pipeline will be created.
    b. If needed, create a new package.
    c. In the **Packages** area, click **+**.
    d. Enter a unique name for the package.
    e. **Optional:** Add a description.
    f. Click **Save**.
    g. Select the new package from the list to continue.
3. Create a new pipeline:
    a. In the **Pipelines** area, click **+ PIPELINE**.
    b. Enter a unique name for the pipeline. For example, `Encrypt Student Worker Data Pipeline`.
    c. **Optional:** Add a description. For example, `This pipeline extracts student worker data from Banner, formats it as a CSV file, encrypts it using PGP encryption, and securely transfers the encrypted file to an SFTP server.` d. Click **Save**.
    e. Click the pipeline name to open it.
    f. To set up parameters, near the pipeline name, click the drop-down menu and select **Edit Pipeline Settings**.
    g. Add the following new pipeline parameters:
        a. publicKeyEncrypt
            - Label: `PGP Public Key`
            - Description: `The PGP public key used to encrypt the student worker data file.`
            - Type: **String**
            - Required: **True**
            - Sensitive: **True**

- Multiline: **True**
- Persistent: **True**

b. privateKeyEncrypt

- Label: `PGP Private Key for Signing`
- Description: `The PGP private key used to sign the encrypted file for authenticity verification.`
- Type: **String**
- Required: **True**
- Sensitive: **True**
- Multiline: **True**
- Persistent: **True**

c. passphraseEncrypt

- Label: `PGP Passphrase`
- Description: `The passphrase associated with the PGP private key used for signing.`
- Type: **String**
- Required: **True**
- Sensitive: **True**
- Persistent: **True**

d. privateKeySFTP

- Label: `SFTP Private Key`
- Description: `The private key used for authenticating with the SFTP server.`
- Type: **String**
- Required: **True**

- Sensitive: **True**
- Multiline: **True**
- Persistent: **True**

e. sftpUsername

- Label: `SFTP Username`
- Description: `The username for authenticating with the SFTP server.`
- Type: **String**
- Required: **True**

f. sftpHost

- Label: `SFTP Host`
- Description: `The hostname or IP address of the SFTP server.`
- Type: **String**
- Required: **True**

g. sftpFilePath

- Label: `SFTP Destination File Path`
- Description: `The full file path on the SFTP server where the encrypted file will be uploaded.`
- Type: **String**
- Required: **True**

h. Click **Save**.

4. To build the first segment for this example, configure the Extract Data fitting:

   a. from the **Insights Transformations** menu, click **Extract Data**.

   b. Enter `Extract Data - Extract Student Worker Data from`

`Banner` as the fitting name.

c. **Optional:** Enter a description. For example, `Extracts active student worker records from Banner, including employee details, position information, and salary data.`

d. Enter the SQL inquiry that will collect the active student worker data from the Banner database.

   This pulls the active student worker data, For example, name, salary, position, employment status. For this example, enter the following SQL statement.

```
SELECT spriden_id,
       NBRJOBS_ORGN_CODE_TS,
       PEBEMPL_ORGN_CODE_HOME,
PEBEMPL_ORGN_CODE_DIST,
nbrjobs_posn,        NBBPOSN_GRADE,
       NBBPOSN_TITLE,
       NBBPOSN_EXEMPT_IND,
       NBBPOSN_POSN_REPORTS,
NBRJOBS_SAL_GRADE,
nbrjobs_ann_salary,
spriden_last_name,
spriden_first_name,
nbrjobs_suff,        nbrjobs_ecls_code
FROM nbrjobs a,
NBBPOSN,
spbpers,
pebempl,      spriden
WHERE nbrjobs_pidm = spriden_pidm
AND spriden_pidm = pebempl_pidm
  AND spriden_pidm = spbpers_pidm
  AND nbrjobs_posn = nbbposn_posn
  AND spriden_change_ind IS NULL
  AND a.nbrjobs_status <> 'T'
  AND a.nbrjobs_ecls_code IN ('US', 'ST', 'GB', 'NR')
  AND nbrjobs_effective_date = (
        SELECT MAX(nbrjobs_effective_date)
        FROM nbrjobs b
        WHERE nbrjobs_effective_date <= now()
AND b.nbrjobs_pidm = a.nbrjobs_pidm
          AND b.nbrjobs_posn = a.nbrjobs_posn
          AND b.nbrjobs_suff = a.nbrjobs_suff
      )
ORDER BY spriden_last_name,
spriden_first_name,
         a.nbrjobs_posn;
```

e. Set **Bulk Load Enabled** to **True**.

f. Set **Ignore Errors** to **False**.

g. Click **Save**.

5. To specify the path to where to extract the data from before you convert the data into a CSV file, configure the Delimited Formatter fitting:

   a. From the **Transformations** menu, select Delimited Formatter.

   b. Enter `Delimited Formatter - Format Data as CSV` as the fitting name.

   c. **Optional:** Enter a description. For example, `Formats the extracted student worker data into a CSV structure with selected columns including name, salary grade, annual salary, position code, and title.`

   d. Set **Normalize Columns** to **False**.

   e. Turn **Column Headers** to **True**.

   Column paths from the Extract Data transform display in the **Columns** field.

```
[
            {
              "path": "spriden_first_name"
            },
            {
              "path": "spriden_last_name"
            },
            {
              "path": "nbrjobs_sal_grade"
            },
            {
              "path": "nbrjobs_ann_salary"
            },
            {
              "path": "nbrjobs_posn"
            },
            {
              "path": "nbbposn_title"
            }
          ]
```

   f. Click **Save**.

   Result: The data is now formatted to write to a CSV file.

6. To write that formatted data to the CSV file, configure the File Writer fitting:

   a. From the **Destinations** menu, select **File Writer**.

b. Enter `File Writer - Write formatted student worker data to a csv file` as the fitting name.

c. Enter a **File Name**. For example, `studentWorkers.csv`.

d. Set **Add New Line After Each Message** to **True**.

e. Set **Include Total Count** to **False**.

f. Set **Generate Presigned Url** to **True**.

g. Set **Write Immediately** to **False**.

h. Set **UTF Encoding** to **UTF-8**.

i. Click **Save**.

7. Configure the File Reader fitting:

   a. From the **Sources** menu, select **File Reader**.

   b. Enter `File Reader - Read the file from S3` as the fitting name.

   c. Set **Key** to {{context._pipelineName}}/{{context._runId}}/studentWorkers.csv. d. Click **Save**.

   Result: The File Reader reads the message and places it into a single payload, as the Encrypt Data fitting requires a single payload. It is recommended when there are multiple messages that you use a Reducer fitting to place the message into a single payload.

8. Configure the Encrypt Data fitting:

   a. From the **Transformations** menu, select **Encrypt Data**.

   b. Enter `Encrypt Data - Encrypt retrieved data` as the fitting name.

   c. Select the **Public Key** parameter `publicKeyEncrypt` that you set up in the pipeline parameters earlier from the drop-down list.

   d. If **Signature** is set to **True**, the **Private Key** becomes required.

   e. **Optional:** Enter `privateKeyEncrypt` for the **Private Key** and `passphraseEncrypt` for the **Passphrase**.

f. Leave the **Binary** setting as the default setting **True** to return encrypted data in binary format.

g. Click **Save**.

9. Configure the SFTP Put fitting:

   a. From the **Destinations** menu, select SFTP Put.

   b. Enter `SFTP Put - Upload Encrypted File to SFTP` as the fitting name.

   c. **Optional:** Enter a description. For example, `Securely uploads the encrypted GPG file to the configured SFTP server using private key authentication`.

   d. Enter `privateKeySFTP` for the **Private Key**.

   e. Enter `{{context.sftpUsername}}` for the **User Name**.

   f. Enter `{{context.sftpHost}}` for the **Host**.

   g. Enter `{{context.sftpFilePath}}` for the **File Path**.

   h. Set **Push** to **True**.

   i. Click **Save**.

Results

The pipeline generates a PGP-encrypted .gpg file containing the student worker data in CSV format. The file is uploaded to the specified SFTP location with the configured filename.